Building this multimodal system has been a real learning curve. It's one thing to have a theory about how vision models and language models should talk to each other, but it's another thing entirely when you actually try to run them both on a laptop or a cloud instance. Today was mostly about figuring out how to stop the models from fighting over the GPU and trying to make sense of the weird things the AI says when it gets confused.
I spent a lot of time troubleshooting before I figured out that I just needed to "hide" one GPU from Ollama using environment variables. It was a good lesson that sometimes "less is more" when it comes to hardware stability is way more important than just having raw power.
Then there were the hallucinations. Seeing YOLO call Napoleon's hat an umbrella was funny, but then the VLM started calling random things an "urn of water" on a building. It really made me realize that these models aren't seeing the world like we do; they're just guessing based on patterns. I had to play around with the temperature settings to make the AI more "boring" so it wouldn't make things up. It's a weird trade-off where you have to turn down the AI's creativity to make it actually useful for a safety project.
I also had to change my whole plan for which models to use. I wanted to use LLaVA for everything, but it was just too slow on my laptop. I ended up with this "half-baked" idea of using a smaller model called Moondream to do the quick frame descriptions and then letting LLaVA read all those notes at the very end to write a summary. It actually works much better this way because the video doesn't lag as much, and the final summary still looks professional.
Looking back, I keep thinking about how I could make this even better. Right now the code is kind of "stop and go" because the video processing has to wait for the AI to finish its sentence. In a perfect world, I'd have the AI running in the background on its own thread so the video never slows down. That's probably what I'll look into next. Today was basically a long session of debugging the "AI's brain" and finding a balance between what's cool and what actually stays running without crashing.